Suggested Teaching Guidelines for
AI Compute Platforms, Applications & Trends
PGCP-AI February 2026

Duration: 60 hours Theory + 60 hours Lab + 60 Hours Self-Learning

Objective: The objective of this module is to make students aware with the latest AI tools and Techniques used by IT Industry

Prerequisites: Knowledge of fundamentals of programming and basic mathematics. ML & statistical knowledge

Evaluation method: Theory exam– 40% weightage
Lab exam – 40% weightage
Internal exam – 20% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share some coursematerials

Reference:

- Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Neural Networks and Learning Machines, Simon Haykin
- Pattern Recognition and Machine Learning, Christopher M. Bishop
- Cloud Computing: Master The Concepts, Architecture and Applications With Real-World Examples And Case Studies by Mehul Mahrishi Kamal Kant Hiran, Ruchi Doshi, Dr. Fagbola Temitayo
- Kubernetes: Up and Running - Dive into the Future of Infrastructure, Second Edition [Paperback] Brendan Burns; Kelsey Hightower and Joe Beda by Brendan Burns (Author), Kelsey Hightower (Author), Joe Beda (Author)
- Hands-on Pipeline as Code with Jenkins: CI/CD Implementation for Mobile, Web, and Hybrid Applications Using Declarative Pipeline in Jenkins (English Edition) by Ankita Patil Mitesh Soni
- Introducing GitHub 2e by Brent Beer

Note:

- **74 hours for PySpark including theory, lab and self-learning sessions.**
- **106 hours for cloud computing including theory, lab and self-learning sessions.**
- **Applications/case studies are advised to be covered under Lab Hours.**
- **Faculty is advised to give appropriate assignment for sufficient hands-on.**
- **Faculty is advised to give at-least one end-to-end problem statement where students can execute their learning in a proper manner.**

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PGCP-AI February 2026

PySpark (28 Hrs of Theory + 22 Hrs of Lab + 24 Hrs of Self-Learning) 74 Hours

Session 1 & 2: (6T + 4L + 3SL)
Theory:

- Apache Spark APIs for large-scale data processing
- Overview, Linking with Spark,
- Initializing Spark
- Spark session & spark context
- Resilient Distributed Datasets (RDDs), External Datasets, RDD Operations,
- Passing Functions to Spark, Working with Key-Value Pairs, Shuffle operations,
- RDD Persistence, Removing Data, Shared Variables, Deploying to a Cluster
- Spark Data Frames

Lab Assignments:

- Run the provided Hadoop Streaming program using python
- Install Spark

Self-learning:

- Reflect on why single-machine data processing breaks down when data size and computation scale increase
- Study Spark's execution flow (driver → executors → tasks) to understand where and why performance bottlenecks occur
- Compare RDDs and DataFrames from a design perspective rather than an API perspective
- Think through why lazy evaluation is beneficial in distributed systems and how it affects debugging

Session 3 & 4: (6T + 4L + 3SL)
Theory:

- EDA using PySpark
- ETL jobs using sparks

Lab Assignments:

- Explore Data using PySpark

Self-learning:

- Observe how exploratory analysis changes when data no longer fits in memory
- Think of ETL as a data pipeline problem instead of a coding problem
- Examine how poor schema understanding leads to incorrect analysis at scale

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PGCP-AI February 2026

Session 5: (4T + 4L +3SL)
Theory:

- Introduction to Kafka
- Working with Kafka using Spark
- Spark streaming

Lab Assignments:

- Install Kafka
- Execute the spark streaming with Kafka

Self-learning:

- Study why batch analytics is insufficient for real-time decision making
- Understand Kafka's role as a durable, distributed message log rather than just a messaging tool
- Visualize and document the life cycle of a streaming record from producer to Spark processing

Session 6: (4T + 2L + 3SL)
Theory:

- Integration of setting Spark and Kafka
- Setting up Kafka Producer and Consumer
- Kafka Connect API

Lab Assignments:

- Connect the Kafka with API

Self-learning:

- Compare manual producer–consumer integration with connector-based ingestion approaches
- Analyze why Kafka Connect is widely used in enterprise data pipelines
- Identify common failure points in streaming integrations and how systems recover

Session 7: (2T + 2L + 3SL)
Theory:

- Machine Learning using Spark's MLlib

Lab Assignments:

- Explore MLib

Self-learning:

- Reflect on why traditional ML workflows do not scale well for very large datasets
- Understand the idea of ML pipelines and staged transformations in Spark
- Decide when Spark MLlib is an appropriate choice and when it is unnecessary overhead

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Session 8: (2T + 2L + 3SL)
Theory:

- Deep Learning using Spark

Lab Assignments:

- Explore Deep Learning

Self-learning:

- Examine Spark's role in deep learning ecosystems primarily as a data processing layer
- Understand the limitations of Spark for model training compared to specialized DL frameworks

Session 9: (2T + 2L + 3SL)
Theory:

- Spark SQL

Lab Assignments:

- Run the Spark SQL programs using step –runs for each and every code line
- Analyze the election data using spark and provide analysis

Self-learning:

- Study why SQL remains relevant even in large-scale distributed systems
- Compare query execution in Spark SQL with execution in traditional RDBMS

Session 10: (2T + 2L + 3SL)
Theory:

- Connecting DB's with Spark
- Working with DB's

Lab Assignments:

- Connecting DB's with Spark
- Accessing and manipulating the DB's
- Create a complex workflow using bash operator, a simple workflow using python

Self-learning:

- Think through challenges involved in moving data from transactional systems into analytics platforms
- Understand why automation is essential for reliable, repeatable data workflows
- Compare ad-hoc execution with scripted and automated pipelines

Suggested Teaching Guidelines for
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PGCP-AI February 2026

DevOps for AI/ML (32 Hrs of Theory and 38 Hrs of Lab +36 Hrs of Self-Learning) 106 Hours

Session 11 & 12: Git / GitHub (4T + 4L + 5SL)
Theory:

- Introduction to Version control systems
- Creating GitHub repository
- Using Git – Introduction to Git commands

Lab Assignment:

- Create a GitHub repository from the command line

Self-learning:

- Creating feature/bugfix branches and understand Merging strategies (fast-forward vs merge commit)
- Understanding merge conflicts and Resolving conflicts manually
- Create clean pull requests, review other developers' code, and respond to review comments while using GitHub or GitLab
- Learn how to use Git rebase
- Learn to safely recover mistakes

Session 13 & 14: Introduction to containers (4T + 6L + 5SL)
Theory:

- Introduction to DevOps
- Introduction to Containers
- Advantages of using container-based applications
- Installing Docker and using basic Docker commands
- Build your own container-based application image
- Networking in Docker
- Managing containers – Logs / Resources

Lab Assignments:

- Create an httpd container running on port 8000. Name the container as web5. Do not map any directory. Create an "index.html" file & copy this file inside the container and check if the website displays your webpage. Again, modify the "index.html" file and copy it inside the container, and check if the website is updated or not.
- Test any program on-base machine. Then create an image using Dockerfile to execute this exe program. Give image name as "Name: v1".

Self-learning:

- Understand DevOps fundamentals and how Docker enables CI/CD, faster deployments, and team collaboration

- Understand containers vs. VMs, Docker architecture, and why containers are favored in modern apps
- Learn the benefits of container-based apps: portability, environment consistency, scalability, and efficient resource use
- Install Docker and practice commands to run, manage, and remove containers
- Create a Dockerfile and build a custom image to understand containers
- Learn Docker networking, including port mapping and container communication internally and externally
- Inspect running containers, view logs, monitor resource usage, try troubleshooting

Session 15 & 16: Introduction to Kubernetes (4T + 6L + 5SL)
Theory:

- Need for Kubernetes
- Introduction to Kubernetes cluster – Basic terms - Management node, Worker Nodes, Pods, Deployment, and Service Types etc.
- Working with Kubernetes Cluster – Creating deployment, Exposing Deployment as a service, managing your applications. Rolling application updates etc.

Lab assignment:

- Create a Kubernetes cluster, deploy and explore any app.

Self-learning:

- Understand the need for Kubernetes and explain how it helps manage, scale, and self-heal containerized applications in production
- Study Kubernetes cluster architecture and core components including control plane, worker nodes, pods, deployments, and common service types
- Learn to work with a Kubernetes cluster by creating deployments, exposing them as services, and managing application replicas
- Understand rolling updates and rollbacks, and practice updating application versions without downtime

Session 17 & 18: CI/CD with Jenkins (4T + 6L + 5SL)
Theory:

- Introduction to CI/CD
- Using Jenkins to build a CI/CD pipeline.

Lab Assignments:

- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the NGINX image.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the httpd image.

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PGCP-AI February 2026

Self-learning:

- Understand the basics of CI/CD and explain how it helps automate building, testing, and deploying AI/ML models or applications
- Learn to use Jenkins to create a simple CI/CD pipeline for AI/ML workflows, including automated code testing, model training, and deployment
- Practice integrating version control (Git) with Jenkins to trigger builds on code
- Understand pipeline concepts like stages, jobs, and artifacts, and explore rolling back or updating ML models safely in production

Cloud Computing:

Session 19 & 20: (4T + 4L +4SL)
Theory:

- Cloud Computing Basics, Understanding Cloud Vendors (AWS/Azure/GCP), Definition, Characteristics, Components

Lab Assignments:

- Study about cloud and other similar configuration
- Explore available solutions
- Cloud Architecture

Self-learning:

- Understand cloud computing basics, its key characteristics, components, and explore major cloud vendors like AWS, Azure, and GCP
- Study cloud architecture, explore available solutions, and practice simple configurations to understand deployment and resource management

Session 21 & 22: (4T + 4L +4SL)
Theory:

- Cloud provider, SAAS, PAAS, IAAS and other Organizational scenarios of clouds, Administering & Monitoring cloud services, benefits, and limitations

Lab Assignments:

- Provide a solution on cloud as SAAS using available systems.

Self-learning:

- Understand cloud service models (SaaS, PaaS, IaaS), their use cases in organizations, and learn the benefits and limitations of each
- Study administering and monitoring cloud services, and practice providing a simple SaaS solution using available cloud platforms

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Session 23 & 24: (4T + 4L +4SL)
Theory:

- Deploy application over cloud. Comparison among SAAS, PAAS, IAAS, Cloud Products and Solutions, Cloud Pricing, Compute Products and Services, Elastic Cloud Compute, Dashboard.

Lab Assignments:

- Deploy any application over the cloud platform.

Self-learning:

- Deploy applications over cloud platforms and explore different deployment scenarios for AI/ML workloads
- Understand and compare cloud service models (SaaS, PaaS, IaaS), cloud products, solutions, pricing, and compute services like Elastic Cloud Compute
- Practice using cloud dashboards to manage resources, monitor deployments, and optimize applications for AI/ML tasks

Session 25 & 26: (4T + 4L +4SL)
Theory:

- Exploring cloud services for AI/ML

Lab Assignments:

- Explore the different cloud computing platforms for AI/ML.

Self-learning:

- AI applications in Financial Services including Insurance banking, stock markets & other financial markets like Forex–and Artificial Economics
- AI applications in Health Sciences & other Scientific Applications, AI in Cloud Environment, Deployment of Models on a distributed platform.